

AMARA DINESH KUMAR

Senior AI Engineer | Agentic AI | GraphRAG | Knowledge Graphs | RAG | AWS

Hyderabad, Telangana, India • 12+ Years Technology Experience • 8+ Years AI/ML • Python • LangGraph • LangChain • LlamaIndex

PROFESSIONAL SUMMARY

Senior AI/ML engineer with 12+ years of technology experience and 8+ years in AI/ML, specializing in production-oriented **Agentic AI, multi-agent orchestration, RAG, GraphRAG, knowledge graphs, LLM evaluation, guardrails and cloud-native AI engineering**. Strong hands-on Python/software engineering background with experience designing reusable agent workflows, tool/function calling, routing, state management, enterprise API integrations, retrieval pipelines, evaluation frameworks and responsible execution controls. Healthcare AI experience at **Evernorth Health Services** includes enterprise RAG, multi-agent systems and a **GraphRAG / Knowledge Graph solution** for improving enterprise grounding and relationship-aware retrieval. Experienced with **LangGraph, LangChain, LlamaIndex, AWS Bedrock, SageMaker, FastAPI, Docker, Kubernetes, Terraform, CI/CD, LangSmith/Langfuse and OpenTelemetry**.

CORE FIT FOR ALPHA INTELLIGENCE

- **Agent Engineering:** Single-agent and multi-agent orchestration, planning, routing, tool/function calling, memory, state management, checkpointing, retries, failure recovery, human-in-the-loop controls
- **GraphRAG / Knowledge:** Knowledge graphs, ontology/semantic modeling concepts, entity and relationship extraction, graph traversal, hybrid retrieval, vector search, grounding, provenance, source attribution and GraphRAG evaluation
- **Evaluation & Observability:** LangSmith, Langfuse, evaluation datasets, retrieval metrics, groundedness, answer quality, regression testing, traces, monitoring and error analysis
- **Responsible AI:** Guardrails, structured outputs, access controls, auditability, privacy-aware enterprise AI patterns, human approvals, prompt/tool controls and governed execution
- **Cloud & Deployment:** AWS Bedrock, SageMaker, S3, Lambda, SQS, Glue, EKS; Docker, Kubernetes, Terraform, CI/CD, REST APIs and production troubleshooting
- **Engineering:** Python, FastAPI, Flask, Git, automated testing, reusable SDK/components, microservices, API integration, performance tuning and documentation

PROFESSIONAL EXPERIENCE

Senior AI/ML Associate | Feb 2026 – Present

JPMorgan Chase

- Engineer enterprise AI/ML solutions spanning LLM applications, retrieval, evaluation and production-oriented AI services.
- Build reusable Python components and APIs for AI workflows, integrating models, tools and enterprise services with reliability and security considerations.
- Apply structured evaluation, guardrails and observability practices to improve reliability of LLM-powered applications.
- Collaborate with engineering and business stakeholders to translate complex requirements into maintainable AI services and measurable outcomes.

Machine Learning Advisor | Jul 2024 – Feb 2026

Evernorth Health Services (Cigna Health Solutions)

- **GraphRAG / Knowledge Graph:** designed and prototyped an enterprise GraphRAG solution combining knowledge graph concepts, entity/relationship modeling, semantic retrieval and LLM generation to improve context-aware enterprise question answering and reasoning.
- Designed graph-aware retrieval flows to complement conventional vector RAG, enabling relationship-based discovery, multi-hop context and more grounded answers from interconnected enterprise knowledge.
- Worked across ingestion, entity/relationship enrichment, retrieval, graph traversal, grounding, source attribution and evaluation for knowledge-intensive AI workflows.
- Built multi-agent RAG and copilot solutions using **LangChain and LangGraph**, applying routing, tool calling, stateful orchestration, guardrails and failure-handling patterns.
- Developed an ExpertGPT Excel Agent with LangChain and guardrails, reducing manual processing/reporting effort by approximately **60%**.
- Worked on contact-center AI automation associated with a **25% improvement in first-response resolution** and optimized targeted AI workflows for approximately **30% accuracy improvement**.
- Implemented evaluation and observability practices using LLM evaluation concepts, traces, retrieval quality analysis, error analysis and production feedback loops.
- Developed Python/FastAPI-based AI services and AWS workflows using S3, SQS, Lambda, Glue and SageMaker, with containerized deployment and CI/CD practices.
- Applied responsible AI controls including prompt/tool guardrails, structured outputs, validation, access-aware workflows and human oversight for sensitive enterprise actions.
- Mentored engineers on agent architecture, RAG, evaluation, guardrails, production reliability and reusable AI engineering patterns.

Research Analyst – AI/ML & R&D; | Aug 2021 – Jul 2024

Toshiba Software Pvt. Ltd. | R&D;

- Developed deep learning, NLP, Transformer and generative AI solutions using Python, PyTorch, TensorFlow and Hugging Face.
- Designed knowledge-intensive RAG systems using embeddings, vector databases, hybrid retrieval and reranking, with focus on grounding and retrieval evaluation.
- Worked on LLM fine-tuning using LoRA, QLoRA and PEFT, and experimented with RLHF/DPO-style alignment approaches.
- Converted research concepts into reusable Python components and technical prototypes with production-oriented software engineering practices.

IT Analyst | Oct 2014 – Aug 2021

Tata Consultancy Services

- Built and supported enterprise software and data solutions across client environments, developing strong foundations in software engineering, APIs, data processing and production delivery.
- Developed Python-based automation, data-processing and ML workflows and collaborated with cross-functional teams through requirements, implementation, testing and production support.

SELECTED AGENTIC AI / GRAPHRAG PROJECTS

- **Enterprise GraphRAG Knowledge Assistant — Evernorth:** Graph-enhanced RAG architecture combining enterprise knowledge modeling, entity/relationship enrichment, graph traversal and semantic/vector retrieval with LLM generation; focused on grounding, relationship-aware reasoning, provenance and source attribution.

- **Multi-Agent RAG Customer Support:** Multi-agent orchestration with routing, retrieval, tool use, state management, grounding and evaluation for enterprise support workflows.
- **ExpertGPT Excel Agent:** Agentic workflow using LangChain, tools and guardrails to automate enterprise data/reporting tasks; delivered approximately 60% reduction in manual processing effort.
- **LLM Analysis Pipeline:** AWS SQS → Lambda router → S3 prompts/context → analysis Lambda → LLM → S3 persistence, designed as a reusable enterprise processing pattern.
- **RAG SDK:** Modular Python SDK for reusable retrieval components, enabling consistent integration of document processing, embeddings, retrieval and generation capabilities.

TECHNICAL SKILLS

Languages: Python, SQL • **Agent Frameworks:** LangGraph, LangChain, LlamaIndex, AutoGen, DSPy • **LLM/AI:** Transformers, Hugging Face, PyTorch, TensorFlow, LoRA, QLoRA, PEFT, RLHF, DPO • **RAG:** GraphRAG, hybrid retrieval, semantic search, reranking, embeddings, grounding, citations, retrieval evaluation • **Graph:** Knowledge Graphs, graph data modeling, entity/relationship modeling, graph traversal, ontology/semantic-model concepts • **Vector/Search:** FAISS, Chroma, Pinecone, Elasticsearch/OpenSearch • **Cloud:** AWS Bedrock, SageMaker, S3, Lambda, SQS, Glue, EKS; Azure, GCP • **Engineering:** FastAPI, Flask, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD • **Observability/Eval:** LangSmith, Langfuse, MLflow, W&B;., OpenTelemetry, Arize Phoenix

EDUCATION

M.Tech, Automotive Electronics — Amrita Vishwa Vidyapeetham, 2017–2019 | CGPA: 8.6 • **B.Tech, Electronics & Communication Engineering** — JNTU Hyderabad, 2010–2014 | CGPA: 7.84

RESEARCH & TECHNICAL CONTRIBUTION

8 research publications across journals and conferences with **500+ citations**, demonstrating sustained applied research experience and the ability to translate technical concepts into measurable experiments, reusable implementations and engineering outcomes.